



Dr. Avit K. Bhowmik

Environmental Scientist | Climate Solutionist

Born on 27 September, 1986

contact

Risk and Environmental Studies
Karlstad University
Universitetsgatan 2
SE 651 88 Karlstad, Sweden

Tel: +46 (0)54 700 10 44

Fax: +46 (0)54 700 10 44

E-mail: avit.bhowmik@kau.se

Web: <http://avitbhowmik.com/>

Publications: Google Scholar

research areas

Sustainability transformation
Climate action
Social tipping points
Agency and Scale of Action
Geostatistics

technical skills

Programing & Modelling

R, C++, Python, Netlogo and
Unix Shell'

Databases

SQL and PostgreSQL

Geoinformatics

GRASS GIS, Quantum GIS,
ArcGIS, GeoMS, and ERDAS
IMAGINE

Communication and Presentation

HTML, CSS, Markdown and $\text{\LaTeX}$

languages

Bangla: mother tongue
English: level C2
Swedish: level B1
German: level C1
Portuguese: level A1

key achievements

2024 - present

Successful research grant from Vinnova (Swedish Innovation Agency)

Main applicant (PI) of the project "Development of sustainability Indicators to drive innovation for Energy Transition (DIET) in Värmland"(200 000 euro).

2024 - present

Successful research grant from Swedish Energy Agency and Viable Cities

Co-applicant with Arvika Municipality, Sweden, of the project "Climate Neutral Arvika 2030"(500 000 euro).

2024 - present

Successful grant by the Swedish Agency for Economic and Regional Growth (co-financed by European Union)

Co-applicant and major contributor to the work-packages on "Energy Communities" and "Local Energy Leadership" of the project "Local Energy Leadership in Värmland (LOKEN)"(2 M euro).

2022 - present

Advisor to "African Circular Business Alliance"

Advising and assisting African Circular Business Alliance in developing sustainable and circular business models for rapidly reducing greenhouse gas emissions

2022 - present

Successful European Commission Erasmus+ grant

Co-applicant and co-leader of the work-package establishing cooperation between government and academia to develop tools and guides for strengthening policies, of the project grant "Alliance of regional innovation ecosystems based on smart sustainable specialization strategies (ARIES4)" (1.38 M euro).

2020 - present

Advisor to "Sustainability Agenda" of Forshaga and Munkfors municipalities

Advising and assisting Forshaga and Munkfors municipalities in developing disrupting emissions reduction strategies for business and industries to achieve climate neutrality by 2040

2020

PNAS article: Social tipping dynamics for stabilizing Earth's climate by 2050

Co-led the development of the concept "Social Tipping Points" and outlined six key interventions that can contagiously and rapidly transform global societies for climate action. See here for details.

2018 - 2019

Exponential Climate Action Roadmap

Led the research and modelling exercises to lay out the first exponential pathways for climate action to halve greenhouse gas emission every decade from 2020 through 2050. The first output report was highlighted in Global Climate Action Summit, held in San Francisco, September 2018, and an update was released during the UN Climate Action Summit 2019.

2018

The World in 2050

Contributed to shape the narrative and pathways for achieving the Sustainable Development Goals of the "Agenda 2030", led by United Nations Sustainable Development Solutions Network (SDSN), the output reports of which have been regularly presented in United Nations High Level Political Forums.

(Levels: A1/2: Basic user - B1/2: Independent user - C1/2:

Proficient user, according to Common European Framework
of Reference for Languages)

2017	European Research Council (ERC) Advanced Grant Co-applicant and contributed with drafting, reviewing and transformation, social tipping points and geospatial research expertise to a successful ERC Advanced grant proposal, i.e. "Earth Resilience in the Anthropocene (ERA)" (2,500,000 euro), Principal Investigator: Prof. Dr. Johan Rockström.
2016	Arctic Science Ministerial Report Provided expert opinion for spatiotemporal variability modelling of climate and resilience in the Arctic, which shaped the policy framework of the "U.S.-Canada Joint Statement on Climate, Energy, and Arctic Leadership" in the first White House Arctic Science Ministerial Meeting on September 28, 2016.

education

06/2026 (exp.)	Docent (Appointment as a University Reader) Risk and Environmental Studies, Faculty of Arts and Social Sciences	Karlstad University, Sweden
09/2012–12/2015	Doctor of Natural Sciences Quantitative Landscape Ecology, Institute for Environmental Sciences Dissertation: <i>Human and Ecological Impacts of Freshwater Degradation on Large Scales. Development and Integration of Spatial Models with Ecological Models for Spatial-ecological Analyses</i> (supervisor: Prof. Dr. Ralf B. Schäfer) Specialization: Spatial Statistics, Freshwater Eco(toxico)logy, Climate Grade: 1.0 (top ranked) on a scale of 1.0 - 3.0 (1.0 best and 3.0 sufficient)	University of Koblenz-Landau, Germany
09/2010–03/2012	Master of Science in Geospatial Technologies Partners: 1. School of Statistics and Information Management, New University of Lisbon, Portugal, 2. Institute for Geoinformatics (ifgi), University of Münster, Germany and 3. Department of Computer Languages and Systems, University of Jaume I, Spain Dissertation: <i>Evaluation of Spatial Interpolation Techniques for Mapping Climate Variables with low sample density. A case study using a new gridded dataset of Bangladesh</i> (supervisor: Prof. Dr. Ana Cristina Costa) Major: Geostatistics, Geographic Information Science (GIS), Climate Grade: 18 (top ranked) on a scale of 1 - 20 (20 best and 1 worst)	Erasmus Mundus, European Commission
12/2004–10/2009	Bachelor of Science in Urban & Regional Planning Department of Urban & Regional Planning Dissertation: <i>Relocation of the Hazaribagh Tannery: Myth or Reality?</i> (supervisor: Prof. Dr. Md. Shakil Akther) Major: GIS, Environmental and Ecological Planning Grade: 3.79 on a scale of 0-4.0 (4.0 best and 0 worst)	Bangladesh University of Engineering & Technology

professional experience

03/2019–present	Assistant Professor Risk and Environmental Studies 15% teaching and 85% research Karlstad University Representative: Sustainable Development Solutions Network Northern Europe	Karlstad University, Sweden
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01/2018–02/2019	Postdoctoral Researcher & Liaison Future Earth 1. Exponential Climate Action project: Backcasting modelling of climate action strategies to achieve Paris Agreement following the Carbon Law 2. The World in 2050 project: Geospatial modelling and spatiotemporal analyses of drivers and processes underlying Sustainable Development Goals 3. Secretariat Liaison: Global Research Projects: (i) Analysis, Integration and Modelling of the Earth System (AIMES) and (ii) Integrated History and Future of People On Earth (IHOPE), and Knowledge Action Networks: (i) Decarbonization and (ii) Sustainable Development Goals	Royal Swedish Academy of Sciences
01/2018–02/2019	Guest Researcher Stockholm Resilience Centre (Planetary Boundaries)	Stockholm University, Sweden
01/2016–12/2017	Postdoctoral Researcher Stockholm Resilience Centre (Planetary Boundaries) Cross-scale dynamics and spatial resilience patterns in social-ecological domains (led by Prof. Dr. Johan Rockström)	Stockholm University, Sweden
01/2016–03/2018	Guest Lecturer Courses: 1. GIS Application, 2. GIS Projects for Advanced, 3. Introduction to open-source software and R for spatial ecotoxicological analyses, and 4. Spatial(-Temporal) Interpolation with R	University of Koblenz-Landau, Germany
11/2015–12/2015	Research Assistant Quantitative Landscape Ecology, Institute for Environmental Sciences Project - AufLand: Understanding effects of land use on the energy transfer across the freshwater-land interface	University of Koblenz-Landau, Germany
04/2012–07/2012	Research Assistant Institute for Geoinformatics (ifgi) Development of online demonstrators and web tools for core concepts of spatial information	University of Münster, Germany
03/2008–10/2010	Project Assistant <i>The Struggle for Urban Livelihoods and the Quest for a Functional City Reconciling Informal and Statutory Planning Institutions in Dhaka, Bangladesh</i> , supported by the German Research Foundation (DFG) and conducted by the Department of Urban and Regional Planning, Faculty of Spatial Planning Preparation of Geographic Information Systems (GIS) databases, analyses and processing of empirical data, conduction of empirical field investigation, i.e. key informant interviews, group discussion, household interviews and exercise of Venn-diagrams and observations	Technical University of Dortmund, Germany

research activities

research visits

03/2018–04/2018	Visiting Scholar Energy Program Regional disparities among the trade-offs and co-benefits in the underlying processes of Sustainable Development Goals	International Institute for Applied Systems Analysis (IIASA)
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research grants & projects

2024	Development of sustainability Indicators to drive innovation for Energy Transition (DIET) in Värmland PI: Avit Bhowmik, Karlstad University Vinnova (Swedish Innovation Agency) Grant: 2024-03767 Total grant: 200 000 euro Main applicant (PI) and Project Leader
2024	Climate Neutral Arvika 2030 Co-applicant with Arvika Municipality, Sweden Swedish Energy Agency and Viable Cities Grant:P2024-02116 Total grant: 500 000 euro Principal Scientist of the project
2022	Alliance of regional innovation ecosystems based on smart sustainable specialization strategies (ARIES4) Coordinator: Universidad Pública de Navarra (UPNA) European Commission Erasmus+ Grant: 101056369 Total grant: 1.38 M euro Contribution: Work-package establishing cooperation between government and academia to develop tools and guides for strengthening policies
2021	MIRAI2.0 – Joint seed funding of Japan-Sweden collaborative projects PIs: Avit Bhowmik and Shunsuke Managi Total grant: 13,400 euro Startup grant for the research project “Marketing technology (MARTECH) for customer behavioral transformation to rapidly reduce food wastes”
2020	Faculty of Arts and Social Sciences Research Advancement Grant PI: Avit Bhowmik Total grant: 36,000 euro Startup grant for research projects
2018	How to develop cross-scale and target seeking transformation scenarios through participatory processes? PI: Avit Bhowmik Bolin Centre for Climate Research Total grant: 9,000 euro Seed grant for pilot research projects on Sustainable Development Goals
2018–present	Exponential Climate Action Roadmap Collaborator: Exponential Roadmap Initiative, IKEA, Finnish Innovation Fund, Mission 2020
2017	Social-ecological Resilience in the Anthropocene PI: DAvit Bhowmik Department of Biology, Stockholm University Total grant: 4,200 euro Supported and supervised three master’s theses projects
2016	Earth Resilience in the Anthropocene (ERA) Co-applicant, PI: Prof. Dr. Johan Rockström European Research Council (ERC) Advanced Grant Total grant: 2,500,000 euro Contributions: (a) wrote and reviewed the grant proposal, (b) led work packages 2 & 4, (c) added geospatial research expertise
2016–present	The World in 2050 Collaborator: Sustainable Development Solutions Network (SDSN), International Institute for Applied Systems Analysis (IIASA)

2012–2015	Quantitative Landscape Ecology Ph.D. Grant University of Koblenz-Landau, Germany German Research Foundation (DFG) supported project Total grant: 36,000 euro	PI: Avit Bhowmik
2010–2012	Erasmus Mundus Scholarship <i>Master of Science in Geospatial Technology</i> The Education, Audiovisual and Culture Executive Agency (EACEA), European Commission Total grant: 24,000 euro	

models & software development

10/2012	ATRIC Automated Accumulation Threshold selection and Riparian Corridor delineation Developed by co-interfacing R and GRASS GIS Development page: https://github.com/AvitBhowmik/ATRIC	
08/2011	SSTP Spatially Shifting Temporal Points Developed using the gstat package utilities in R Development page: https://github.com/AvitBhowmik/SSTP	

administrative positions and duties

01/2018–present	Research Director Centre for Sustainable Societal Transformation	Karlstad University
03/2019 - present	Course Director and Developer CCGA08: Nordic Climate Change Studies CCGA01: Sustainable Development with a focus on Climate Change RHG310: Climate Change Adaptation and Mitigation MVG300: Environment, Risk and Sustainable Development	Karlstad University
01/2018–present	Lead Researcher Lead partners: IKEA, WWF, Finnish Innovation Fund, Mission 2020	Exponential Roadmap Initiative
09/2021	Principal Organizer Book Seminar: Under the Sky We Make. How to be Human in a Warming World by Dr. Kimberly Nicholas from Centre for Sustainability Studies, Lund University Centre for Sustainable Societal Transformation, Karlstad University, Sweden	
04/2018	Principal Organizer Workshop: Exponential Climate Action for Cities, Carbon Law Accelerator project, Future Earth, Stockholm, Sweden	
02/2018	Theme Leader Pathways to new collaborations in climate research at Stockholm University: Perspectives from the Human Science Academic Area and the Bolin Centre for Climate Research	
08/2017	Session Chair Session: Exploring the potential of local innovations and bottom-up change, Social-ecological transformations for sustainability, Resilience 2017 - Resilience Frontiers for Global Sustainability	

06/2017	Principal Organizer Workshop: Progress in computational macroecology, The 17th International Conference on Computational Science and Its Applications (ICCSA 2017), Trieste, Italy
12/2016	Principal Organizer Social Tipping Elements Instrumental for Decarbonization by 2050: an Expert Workshop, The Earth League, Stockholm Resilience Centre, Sweden
09/2016	Co-organizer Climate Tipping Points and Safe Pathways to Sustainable Development: an expert review workshop, Stockholm Resilience Centre, Stockholm, Sweden Analysis, Integration and Modelling of the Earth System (AIMES)
05/2016	Principal Organizer 3rd Earth-Doc Meeting, Stockholm Resilience Centre, Sweden (a) Preparation of a strategy document outlining Earth-Doc research, and (b) Advancing on the on-going Earth-League research proposal and perspective paper
07/2015	Principal Organizer Workshop 1: Data analysis in freshwater ecology using R, 9th Symposium for European Freshwater Sciences (SEFS), Geneva, Switzerland
09/2012–12/2015	Developer and Maintainer Geodata Server Institute for Environmental Sciences, University of Koblenz-Landau, Germany
11/2010–02/2011	Group Leader Spatial statistics group Group project seminar on Identifying optimal location for disaster evacuation centres in Lisbon, New University of Lisbon, Portugal
07/2007–present	U.S. Department of State Alumni Study of the United States Institute for Student Leaders 2007, organized by the United States Department of State in Washington and Green River Community College, Seattle, USA Community services

editorial activities

2018–2019	Guest Editor Sustainability Science
2016–2019	Reviewer for research project grants Future Earth Sustainable Development Goals Labs Chilean National Science and Technology Commission
2016–present	Editorial board member Water Science and Technology: Water Supply
2014–present	Ad-hoc reviewer Nature: Communications Earth & Environment Nature: npj Urban Sustainability Global Ecology and Biogeography Ecography International Journal of Climatology Sustainability

PLOS ONE
 Earth Science Informatics
 Geomatics, Natural Hazards and Risk
 Journal of Environmental Informatics
 Applied Ecology and Environmental Research
 Journal of Health and Pollution
 Natural Hazards
 Geosciences
 Geoscience Data Journal
 Environmental Monitoring and Assessment
 International Journal of Environmental Research and Public Health
 Agronomy
 Geomorphology

science-policy-industry networks

2022–present	Analysis, Integration and Modelling of the Earth System (AIMES) Scientific Steering Committee Member
2020–present	Forshaga and Munkfors municipalities, Sweden Project: Sustainability Agenda. Development of a Climate Action Framework to achieve carbon neutrality by 2045
2016–2019	Earth-Doc, The Earth League Project: Earth Lander - Social tipping elements instrumental for decarbonization by 2050
2016–2018	European Space Agency Project: Earth System Data Cube, partners: Brockmann Consultant GmbH, Max Planck Institute for Biogeochemistry and Stockholm Resilience Centre
2016–present	Sustainable Development Solutions Network (SDSN) Project: The World in 2050 (TWI 2050)
2015–2020	European Geosciences Union (EGU) Member
2013–2016	Society of Ecotoxicology and Chemistry (SETAC) Member

jury and examination

10/2021	Licentiate Defence Opponent Risk and resilience: An integrated approach for navigating a complex world Emmy Wassénus, Doctoral dissertation Stockholm Resilience Centre, Stockholm University, Sweden and Global Economic Dynamics and the Biosphere, Royal Swedish Academy of Science, Sweden
07/2021	Ph.D. Defence Jury GIS Applications on the Essential Public Services in Mozambique António Dos Anjos Luis, Doctoral dissertation Nova Information Management School, New University of Lisbon, Portugal

publications, presentations, outreach

bibliographic metrics

h-index: 24

i10-index: 30

3111 citations since 2012

source: google scholar

working papers

Equitable sustainability transformation should be informed by history

A. Bhowmik, H. Enefalk, and E. Svensson

under preparation ()

Climate friendly diet as a quasi-religion

A. Bhowmik and K. Plank

under preparation ()

Social tipping points should be the main discourse of climate action

Avit Bhowmik and Teresa Spantzel

under preparation (2020). 2020

Research linking climate change and conflicts should study positive feedbacks

Frederik Wienker and Avit Bhowmik

under preparation (2020). 2020

peer reviewed journal articles

Living beyond limits: Consequences of missing the decisive decade for preserving our planet's life-supporting systems

Johan Rockström, Peter Schlosser, Avit K. Bhowmik, Roger Cremades, Jonathan F. Donges, James G. Dyke, Kristie L. Ebi, Adrian Heilemann, Daniela Jacob, Paula Mirazo, Denise Moreno Ramírez, Nebojsa Nakicenovic, Ilona M. Otto, Franz Prettenhaler, Janet Ranganathan, Hans Joachim Schellnhuber, Rachel Warren, and Henry C. Wu

One Earth (Nov. 2025) p. 101527. 2025. DOI: [10.1016/j.oneear.2025.101527](https://doi.org/10.1016/j.oneear.2025.101527)

Negative social tipping dynamics resulting from and reinforcing Earth system destabilization

Viktoria Spaiser, Sirkku Juhola, Sara M Constantino, Weisi Guo, Tabitha Watson, Jana Sillmann, Alessandro Craparo, Ashleigh Basel, John T Bruun, Krishna Krishnamurthy, Jürgen Scheffran, Patricia Pinho, Uche T Okpara, Jonathan F Donges, Avit Bhowmik, Taha Yasseri, Ricardo Safra de Campos, Graeme S Cumming, Hugues Chenet, Florian Krampe, Jesse F Abrams, James G Dyke, Stefanie Rynders, Yevgeny Aksenov, and Bryan M Spears

(2024). 2024

From Climate Endgame to Climate Long Game

Avit Bhowmik, Mark S. McCaffrey, and Juliette Rooney Varga

Proceedings of the National Academy of Sciences 119.45 (2022) e2214975119. 2022. DOI: [10.1073/pnas.2214975119](https://doi.org/10.1073/pnas.2214975119)

Global Mangrove Deforestation and Its Interacting Social-Ecological Drivers: A Systematic Review and Synthesis

Avit K. Bhowmik, Rajchandar Padmanaban, Pedro Cabral, and Maria M. Romeiras

Sustainability 14.8 (2022). 2022. DOI: [10.3390/su14084433](https://doi.org/10.3390/su14084433)

Powers of 10: seeking “sweet spots” for rapid climate and sustainability actions between individual and global scales

A.K. Bhowmik, M. McCaffrey, A. Ruskey, C. Frischmann, and O. Gaffney

Environmental Research Letters (2020). 2020. DOI: <https://doi.org/10.1088/1748-9326/ab9ed0>

Tipping to Staying on the Ground: Internalized Knowledge of Climate Change Crucial for Transformed Air Travel Behavior

Lisa Jacobson, Jonas Åkerman, Matteo Giusti, and Avit K Bhowmik
Sustainability 2020 (2020) p. 18. 2020. DOI: [10.3390/su12051994](https://doi.org/10.3390/su12051994)

Social tipping dynamics for stabilizing Earth's climate by 2050

Ilona M. Otto, Jonathan F. Donges, Roger Cremades, Avit Bhowmik, Richard J. Hewitt, Wolfgang Lucht, Johan Rockström, Franziska Allerberger, Mark McCaffrey, Sylvanus S. P. Doe, Alex Lenferna, Nerea Morán, Detlef P. van Vuuren, and Hans Joachim Schellnhuber
Proceedings of the National Academy of Sciences (Jan. 15, 2020) p. 201900577. 2020. DOI: [10.1073/pnas.1900577117](https://doi.org/10.1073/pnas.1900577117)

Matching scope, purpose and uses of planetary boundaries science

A. S. Downing, A. K. Bhowmik, D. Collste, S. Cornell, J. Donges, I. Fetzer, T. Häyhä, J. Hinton, S. Lade, and W. M. Mooij
Environmental Research Letters (2019). 2019. DOI: <https://doi.org/10.1088/1748-9326/ab22c9>

Satellite Image Fusion to Detect Changing Surface Permeability and Emerging Urban Heat Islands in a Fast-Growing City

R. Padmanaban, A. K. Bhowmik, and P. Cabral
PLOS ONE 14.1 (2019) e0208949. 2019. DOI: <https://doi.org/10.1371/journal.pone.0208949>

Persistent Organic Pollutant emission via dust fallout throughout Pakistan: Fingerprinting of recent inputs, regional cycling and their implication for human health risks

M. Sohail, J. Podgorski, A.K. Bhowmik, A. Mahmood, N. Ali, H. Bokhari, T. Sabo-Attwood, H. Shen, and S.A.M.A.S Eqani
Science of the Total Environment 618 (2018) pp. 829–837. 2018. DOI: <https://doi.org/10.1016/j.scitotenv.2017.08.224>

Comparison of spatial interpolation techniques to generate high-resolution climate surfaces for Nigeria

Aisha Olushola Arowolo, Avit Kumar Bhowmik, Wei Qi, and Xiangzheng Deng
International Journal of Climatology (Jan. 2017). 2017. DOI: [10.1002/joc.4990](https://doi.org/10.1002/joc.4990)

Modelling Urban Sprawl Using Remotely Sensed Data: A Case Study of Chennai City, Tamilnadu

Rajchandar Padmanaban, Avit K. Bhowmik, Pedro Cabral, Alexander Zamyatin, Oraib Almegdadi, and Wang Shuangao
Entropy 19.4 (Apr. 7, 2017) p. 163. 2017. DOI: [10.3390/e19040163](https://doi.org/10.3390/e19040163)

A Remote Sensing Approach to Environmental Monitoring in a Reclaimed Mine Area

Rajchandar Padmanaban, K. Avit Bhowmik, and Pedro Cabral
ISPRS International Journal of Geo-Information 6.12 (2017). 2017. DOI: [10.3390/ijgi6120401](https://doi.org/10.3390/ijgi6120401)

Human Arsenic exposure via dust across the different ecological zones of Pakistan

Ambreen Alamdar, Syed Ali Musstjab Akber Shah Eqani, Saeed Waqar Ali, Mohammad Sohail, Avit Kumar Bhowmik, Alessandra Cincinelli, Marghoob Subhani, Bushra Ghaffar, Rizwan Ullah, Qingyu Huang, and Heqing Shen
Ecotoxicology and Environmental Safety 126 (Apr. 2016) pp. 219–227. 2016. DOI: [10.1016/j.ecoenv.2015.12.044](https://doi.org/10.1016/j.ecoenv.2015.12.044)

Mercury contamination in deposited dust and its bioaccumulation patterns throughout Pakistan

Syed Ali Musstjab Akber Shah Eqani, Avit Kumar Bhowmik, Sehrish Qamar, Syed Tahir Abbas Shah, Muhammad Sohail, Sikandar I. Mulla, Mauro Fasola, and Heqing Shen
Science of The Total Environment 569-570 (Nov. 2016) pp. 585–593. 2016. DOI: [10.1016/j.scitotenv.2016.06.187](https://doi.org/10.1016/j.scitotenv.2016.06.187)

Spatial distribution of dust-bound trace elements in Pakistan and their implications for human exposure

Syed Ali Musstjab Akber Shah Eqani, Ayesha Kanwal, Avit Kumar Bhowmik, Mohammad Sohail, Rizwan Ullah, Syeda Maria Ali, Ambreen Alamdar, Nadeem Ali, Mauro Fasola, and Heqing Shen
Environmental Pollution 213 (June 2016) pp. 213–222. 2016. DOI: [10.1016/j.envpol.2016.02.017](https://doi.org/10.1016/j.envpol.2016.02.017)

Pesticide mixtures in streams of several European countries and the USA

Verena C. Schreiner, Eduard Szöcs, Avit Kumar Bhowmik, Martina G. Vijver, and Ralf B. Schäfer
Science of The Total Environment 573 (2016) pp. 680–689. 2016. DOI: <http://dx.doi.org/10.1016/j.scitotenv.2016.08.163>

Mapping human health risks from exposure to trace metal contamination of drinking water sources in Pakistan

Avit Kumar Bhowmik, Ambreen Alamdar, Ioannis Katsoyiannis, Heqing Shen, Nadeem Ali, Syeda Maria Ali, Habib Bokhari, Ralf B. Schäfer, and Syed Ali Musstjab Akber Shah Eqani
Science of The Total Environment 538 (Dec. 2015) pp. 306–316. 2015. DOI: 10.1016/j.scitotenv.2015.08.069

Spatially shifting temporal points: estimating pooled within-time series variograms for scarce hydrological data

Avit Kumar Bhowmik and Pedro Cabral

Hydrology and Earth System Sciences Discussion 12 (2015) pp. 2243–2265. 2015. DOI: 10.5194/hessd-12-2243-2015

An automated, objective and open source tool for stream threshold selection and upstream riparian corridor delineation

Avit Kumar Bhowmik, Markus Metz, and Ralf B. Schäfer

Environmental Modelling & Software 63.0 (2015) pp. 240–250. 2015. DOI: 10.1016/j.envsoft.2014.10.017

Large Scale Relationship between Aquatic Insect Traits and Climate

Avit Kumar Bhowmik and Ralf B. Schäfer

PLOS ONE 10.6 (June 16, 2015) e0130025. 2015. DOI: 10.1371/journal.pone.0130025

Representativeness impacts on accuracy and precision of climate spatial interpolation in data-scarce regions

Avit Kumar Bhowmik and Ana Cristina Costa

Meteorological Applications 22 (2014) pp. 368–377. 2014. DOI: 10.1002/met.1463

Industries' Location as Jeopardy for Sustainable Urban Development in Asia: A Review of the Bangladesh Leather Processing Industry Relocation Plan

Avit Kumar Bhowmik

Environment and Urbanization Asia 4.1 (Mar. 1, 2013) pp. 93–119. 2013. DOI: 10.1177/0975425313477749

Temporal Patterns of the Two-Dimensional Spatial Trends in Summer Temperature and Monsoon Precipitation of Bangladesh

Avit Kumar Bhowmik

ISRN Atmospheric Sciences 2013 (2013) pp. 1–16. 2013. DOI: 10.1155/2013/148538

Cyclone Sidr Impacts on the Sundarbans Floristic Diversity

Avit Kumar Bhowmik and Pedro Cabral

Earth Science Research 2.2 (2013). 2013. DOI: 10.5539/esr.v2n2p62

A Comparison of Bangladesh Climate Surfaces from the Geostatistical Point of View

Avit Kumar Bhowmik

ISRN Meteorology 2012 (2012) pp. 1–20. 2012. DOI: 10.5402/2012/353408

A Geostatistical Approach to the Seasonal Precipitation Effect on Boro Rice Production in Bangladesh

Avit Kumar Bhowmik

International Journal of Geosciences 03.3 (2012) pp. 443–462. 2012. DOI: 10.4236/ijg.2012.33048

books and book chapters

1.5°C Business Playbook. Version 2. Building a strategy for exponential climate action towards net-zero emissions.

J. Falk, P. Bergmark, O. Gaffney, G. Erselius, J. Widheden, J. Myrman, M. Jiborn, D. Pamlin, J. Hising, A. K. Bhowmik, and S. Henningson

(Eds), Exponential Roadmap Initiative, 2022

Space-Time Variability of Summer Temperature Field over Bangladesh during 1948-2007

Avit Kumar Bhowmik and Pedro Cabral

Lecture Notes in Computer Science Computational Science and Its Applications – ICCSA 2013, Murgante, Beniamino, Sanjay Misra, Maurizio Carlini, Carmelo M. Torre, Hong-Quang Nguyen, David Taniar, Bernady O. Apduhan, and Osvaldo Gervasi(Eds), 7974, pp. 120–135, Springer Berlin Heidelberg, 2013. ISBN: 978-3-642-39648-9 978-3-642-39649-6

Statistical Evaluation of Spatial Interpolation Methods for Small-Sampled Region: A Case Study of Temperature Change Phenomenon in Bangladesh

Avit Kumar Bhowmik and Pedro Cabral

Lecture Notes in Computer Science *Computational Science and Its Applications - ICCSA 2011*, Murgante, Beniamino, Osvaldo Gervasi, Andrés Iglesias, David Taniar, and Bernady O. Apduhan(Eds), 6782, pp. 44–59, Springer Berlin Heidelberg, 2011. ISBN: 978-3-642-21927-6 978-3-642-21928-3

Application of GIS in Rural Settlement Development. An Approach towards Identification of Locations for Compact Towns

Avit Kumar Bhowmik

Nagar Shoilee, Hafiz, Roxana(Eds), 4, pp. 128–135, Bangladesh University of Engineering & Technology Dhaka, 2009. ISBN: 978-984-33- 0128-4

Environmental concerns regarding Hazaribagh tannery area and present relocation scenario

Avit Kumar Bhowmik and M Samiul Islam

Nagar Shoilee, Hafiz, Roxana(Eds), 4, pp. 33–41, Bangladesh University of Engineering and Technology Dhaka, 2009. ISBN: 978-984-33-0128-4

reports & policy papers

The Amazon Rainforest at the Brink. Global Tipping Points Report.

C.A. Boulton, P. Pinho, A. Bhowmik, S. Constantino, B. M. Flores, M. Hirota, M. Holmgren, I. Parry, T. Powell, J. A. Sanabria-Fernández, V. Spaizer, S. Villasante, and N. Wunderling

A report to COP30 Presidency, Global Systems Institute, University of Exeter, Exeter, 2025

Global Tipping Points Report

A. Carla Staver, A. Israel Muñoz Castillo, Aileen Lam, Alessandro C.W. Craparo, Alexander Robinson, Alexandra Malmström, Alexey Fedorov, Alina Bill-Weilandt, Andreas Born, Andreas Kääb, Aneeqe Javaid, Angeles G. Mayor, Ann Kristin Klose, Anna von der Heydt, Antony Emenyu, Arie Staal, Ashish Ghadiali, Ashleigh M. Basel, Avit Bhowmik, and Azucena Castro

Report to the Conference of the Parties 28, Dubai, 2023

Building back better from the coronavirus disease (COVID-19) while advancing the full implementation of the 2030 Agenda for Sustainable Development

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Karlstad, January 28, 2026

A handwritten signature in black ink, appearing to read 'Avit K. Bhowmik', with a stylized, flowing script.

Avit K. Bhowmik